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Search-Less Algorithm for Star Pattern Recognition[1]

Daniele Mortari[2]

Abstract

A fast and robust star identification technique (which does not use a searching phase) is presented for a spacecraft equipped with a wide field-of-view star tracker. The algorithm, which identifies stars within a large star catalog using stars' angular separation only, consists of two identification processes: the **K**-vector Star-Pair Identification Technique (SPIT), which, for any observed angular separation, selects an admissible star-pair set from the star catalog; and the subsequent Reference-Star Star-Matching Identification Technique (SMIT), which performs the identification process based on a subset of all of the admissible star-pairs. The **K**-vector SPIT, which uses an appropriate devised vector of integers, does not require a searching phase. The best-fitting SPIT, which uses a best-fitting criterion to reduce the searching phase, is also presented. The resulting robustness of the algorithm is such that, after spikes are deleted, at least three true stars are still available. An overall software block diagram of the proposed system is depicted as well as the results obtained by extensive tests.

Introduction

Star observations are widely used by spacecraft as a primary means of attitude determination. Currently, star tracker sensors based on charge coupled devices (CCD) allow us to obtain the best spacecraft attitude estimation because of the high precision data provided. Three-axis attitude estimation algorithms need the knowledge of at least two different directions; therefore, a narrow field-of-view (FOV) star tracker cannot be used in a standalone configuration but must be used together with other less accurate sensors. On the contrary, wide FOV star trackers can be used autonomously and, therefore, provide a high precision dataset to the attitude estimation algorithm. Figure 1 is a sketch of a spacecraft-mounted wide-FOV star sensor.

Obviously, when a wide FOV star tracker is used, speed and reliability of the star identification technique are of capital importance just as speed and accuracy are important for the attitude estimation algorithm. This paper satisfies the above

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[2]Assistant Professor, Aerospace Department, Universita degli Studi "La Sapienza," di Roma, Via Salaria 851, 00138 Rome (Italy). Member AAS. E-Mail: daniele@crarisc1.psm.uniroma1.it.

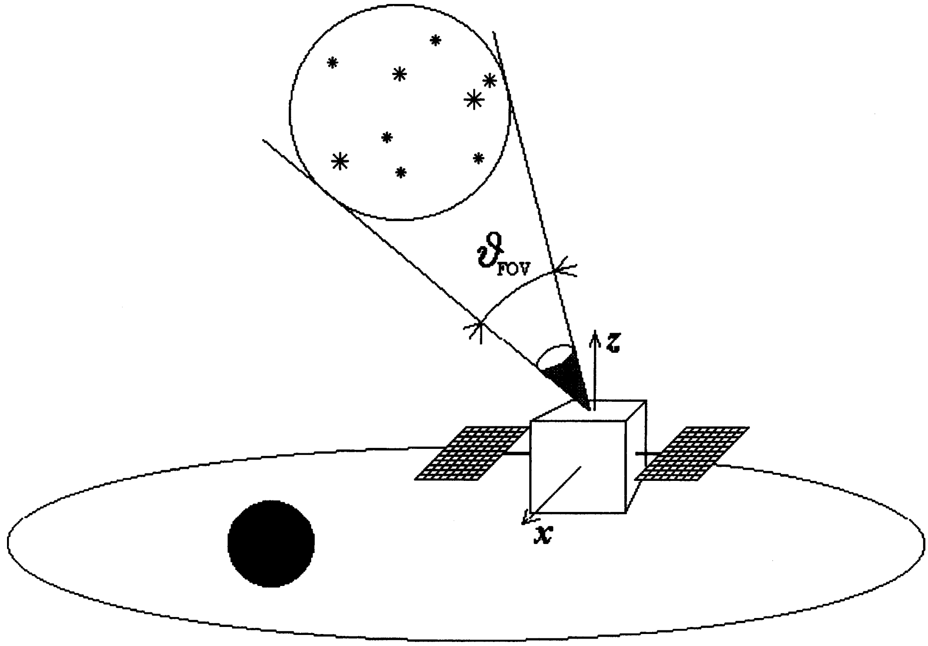

FIG. 1. Sketch of a Wide-FOV Star Sensor.

requirements by proposing the fast and reliable Search-Less Algorithm (SLA) for star pattern recognition which identifies the observed stars, seen by a wide FOV star tracker, within an overall large star catalog. This task is accomplished using stars angular separation only. SLA consists of: 1) the search-less **K**-vector Star-Pair Identification Technique (SPIT), and 2) the Reference-Star Star-Matching Identification Technique (SMIT). The proposed SLA working in conjunction with the ESOQ [1] (currently the fastest available algorithm for optimal attitude estimation) constitutes a fast and robust autonomous attitude determination system.

Although CCDs have improved, the "CCD magnitudes" are still somewhat uncertain and non-standard because various CCDs have different spectral responses. It is also to be noted that the CCD sensitivity may change with time so only relative magnitudes can be relied on. Hence, CCD is best at determining spatial positions so this criteria should be the primary measure for any star identification algorithm. The use of the magnitude information should be restricted to the very few ambiguous cases for the identification technique based on spatial positions only: 1) when only two stars are observed, and 2) in the limit cases when the polygonal pattern of the observed stars (e.g. triangle) cannot be determined univocally within the sensor data precision. In order to avoid these cases, the star sensor FOV ϑ_{FOV} and the set magnitude threshold M_T, are designed to assure a sufficient average number of observed stars. The magnitude range will depend on the lens aperture ϑ_{FOV} and integration times. Typical trackers have 8×8, 20×20, and 32×32 deg FOV and integration times from 0.05 to 1 sec. These values are selected to yield at least 4-5 stars in the FOV (at least 4-5 stars are required to assure a reliable identification if the positions are precise, more are needed if the precision is lower). Thus, for the 8×8 FOV approximately 6th magnitude is needed while for the 20×20 FOV it is enough to detect down to about 4.5-5.0 magnitude and a 2.5-3.5 magnitude for the 32 × 32 FOV. With these limits there may be a few "holes" where there are fewer than 4-5 stars. Let ϵ be the sensor precision. This means that the true star direction falls within a cone about the observed star having angle ϵ.

Another problem is how to accommodate stars that are on the edge of the star camera detection threshold, that is, the missing stars. The catalog contains only magnitude estimates accurate to ± 0.2 magnitude. Since the number of stars is exponential with magnitude (see Fig. 2), 20% of stars in any FOV will lie at the edge of the detection threshold M_T. This means that any star detected has almost a 20% chance of not being in the catalog. However, looking at ways to use the brighter stars in preference over dimmer ones entails the problem of having holes in many directions. On the other hand, the instrument FOV cannot be just of any size.

Figure 3 shows the relationship among the instrument FOV, the magnitude threshold M_T and the average number of observed stars. Therefore, since the accuracy of the magnitude information is often poor and not stable with time, a basic philosophy of the identification technique should use only the information of the stars' angular separations. The use of the magnitude information could, at least, be restricted to the ambiguities only when they actually occur (e.g. when only $n = 2$ stars are observed). However, any star identification technique which does not take advantage of the use of magnitude information depends to a greater extent on the stars direction precision detected by the CCD camera. This precision will become better as the studies on the centroiding methods (how much one should defocus stars to improve the precision of their centroid prediction and how best to weigh the pixels to work out where the center of the star is) are improved. Since

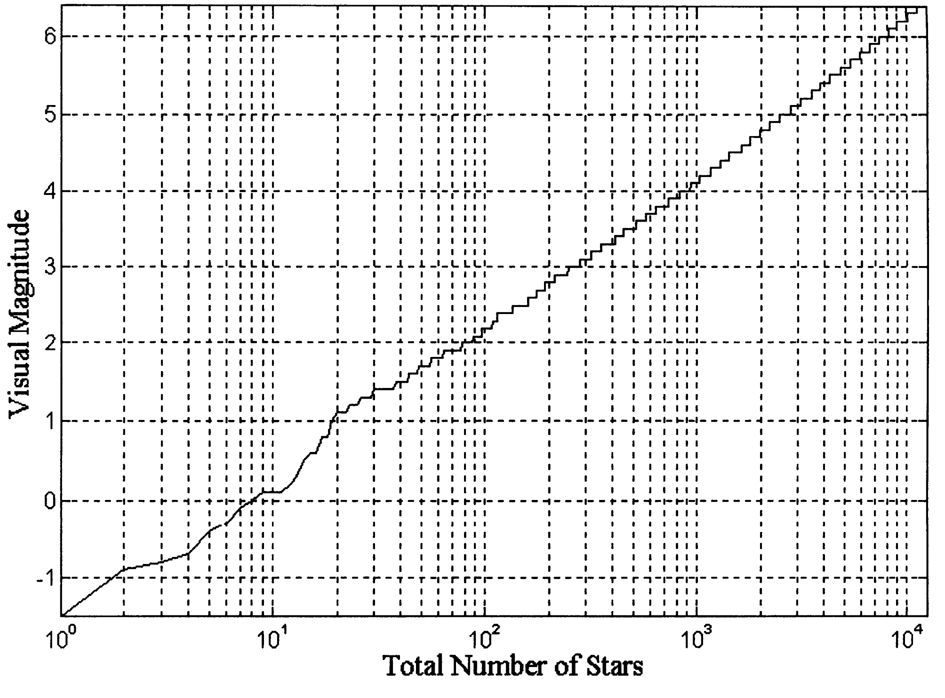

FIG. 2. Total Number of Stars.

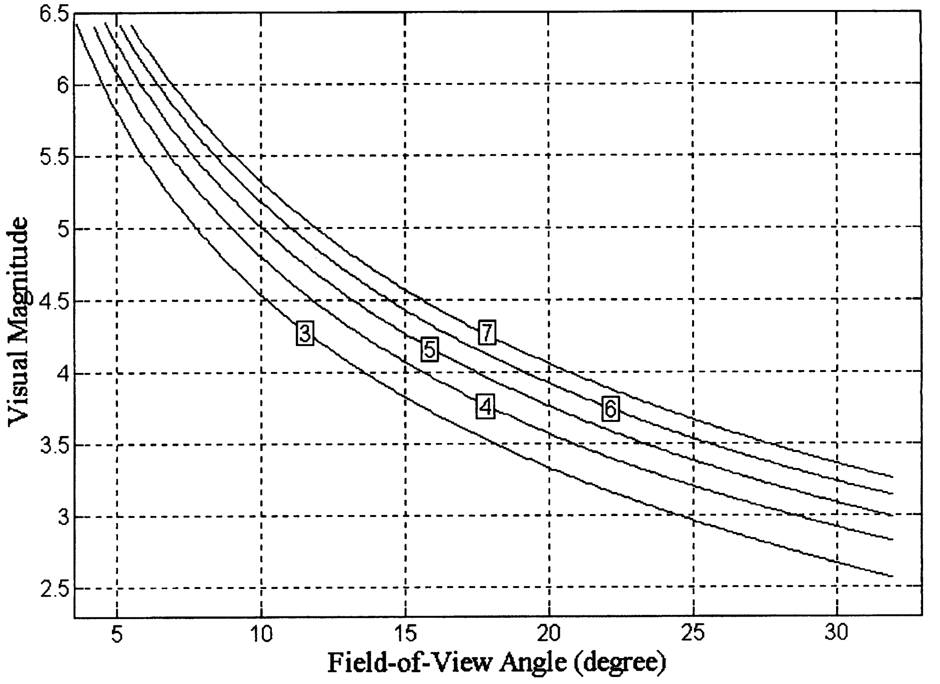

FIG. 3. Average Number of Observed Stars.

the star identification method described in this paper uses only the information of the stars' angular separations, its characteristics will improve with the CCD's camera precision.

All numerical tests performed in this paper refer to a typical star sensor providing a precision of $\epsilon = 10$ arcsec, with a $\vartheta_{\text{FOV}} = 30$ deg FOV aperture and magnitude threshold M_T set to 3. The star catalog used is located in the "*pub/star_catalogs*" directory of the "*s3gsun1.jhuapl.edu*" workstation and is identified as "*MSX_SIM_cat.dat*". This star catalog lists 12,620 stars covering the magnitude range from -1.5 up to 6.4 with a magnitude precision of 0.2. The catalog was not upgraded for the small effects of the stars proper motions as well as the star-light aberration.

The star pattern identification algorithm proposed here consists of two identification processes. The first one, called **K**-vector, identifies a small set of likely catalog star-pairs from the star catalog and for each angular separation of the observed stars. The outputs of this identification will then be the basis of the second identification process, called Reference-Star, which identifies the correspondence between the angular structures of the observed stars (simple separations, triangles) and those built from the list of the likely detected ones by the star-pair identification phase. Two different SPITs (Best-Fitting and **K**-vector) and a SMIT (Reference-Star), which will be explained in the following three sections, are proposed. Comparisons of the SLA algorithm versus existing approaches (angular separations [2-4], angular separation with knowledge of an attitude initial guess [5-7], triangle [8,9], fuzzy logic and neural network [10-12], and stochastic [13]), will be the subject of a future paper.

Star-Pair Identification Technique (SPIT)

Let us consider a star catalog of n stars whose directions are described in the inertial reference system by the unit vectors $\mathbf{v}_i(i = 1 - n)$. If ϑ_{FOV} is the angle indicating the star tracker FOV lens aperture, two stars, $\mathbf{v}_i$ and $\mathbf{v}_j$, might simultaneously be within the instrument FOV, if and only if

$$\mathbf{v}_i^{\mathrm{T}}\mathbf{v}_j \geq \cos\vartheta_{\text{FOV}} \tag{1}$$

Any star-pair satisfying equation (1) is called "admissible" for the sensor FOV. Let $\mathbf{P}$ be the m-dimensional vector containing the dot products of all the admissible star-pairs; that is, $\mathbf{P}(k) = \mathbf{v}_i^{\mathrm{T}}\mathbf{v}_j$, (where $1 \leq k \leq m$ and $1 \leq i \neq j \leq n$). Let $\mathbf{I}_p$ and $\mathbf{J}_p$ be the m-dimensional integer vectors containing, at kth position, the i and j indices, respectively; that is

$$\begin{aligned}\mathbf{P} &= \{\cdots \ \mathbf{v}_i^{\mathrm{T}}\mathbf{v}_j \ \cdots\}^{\mathrm{T}}\\ \mathbf{I}_p &= \{\cdots \quad i \ \cdots\}^{\mathrm{T}}\\ \mathbf{J}_p &= \{\cdots \quad j \ \cdots\}^{\mathrm{T}}\end{aligned} \tag{2}$$

where $\mathbf{I}_p(k) = i$, $\mathbf{J}_p(k) = j$, and $\mathbf{P}(k) = \mathbf{v}_i^{\mathrm{T}}\mathbf{v}_j$ establish the corresponding relationships among $\mathbf{I}_p$, $\mathbf{J}_p$ and $\mathbf{P}$.

Let $\mathbf{S}$ be the sorted vector of $\mathbf{P}$ arranged in ascending order, and $\mathbf{I}$ and $\mathbf{J}$ be, respectively, the associated integer vectors which, so as in the case of $\mathbf{I}_p$ and $\mathbf{J}_p$ with respect to $\mathbf{P}$, maintain the corresponding relationships with $\mathbf{S}$. These vectors

$$\begin{aligned}\mathbf{S} &= \{\cdots \ \mathbf{P}(k) \ \cdots \ \ \mathbf{P}(l) \ \cdots\}^{\mathrm{T}}\\ \mathbf{I} &= \{\cdots \ \mathbf{I}_p(k) \ \cdots \ \mathbf{I}_p(l) \ \cdots\}^{\mathrm{T}}\\ \mathbf{J} &= \{\cdots \ \mathbf{J}_p(k) \ \cdots \ \mathbf{J}_p(l) \ \cdots\}^{\mathrm{T}}\end{aligned} \tag{3}$$

are such that, if $1 \leq i_k \leq i_l \leq m$, then

$$\cos\vartheta_{\text{FOV}} \leq \mathbf{S}(i_k) = \mathbf{P}(k) < \mathbf{P}(l) = \mathbf{S}(i_l) \leq 1 \tag{4}$$

where

$$\begin{matrix}\mathbf{I}(i_k) = \mathbf{I}_p(k) \\ \mathbf{I}(i_l) = \mathbf{I}_p(l)\end{matrix} \quad \text{and} \quad \begin{matrix}\mathbf{J}(i_k) = \mathbf{J}_p(k) \\ \mathbf{J}(i_l) = \mathbf{J}_p(l)\end{matrix} \tag{5}$$

By using the m-dimensional vectors $\mathbf{S}$, $\mathbf{I}$ and $\mathbf{J}$, it is possible to identify, in the vector $\mathbf{S}$, the induces of the admissible star-pairs which could provide the observed dot product $\cos\vartheta$. For the binary searching technique [4] the resulting identification routine may be executed in a time proportional to $L = \log_2(m)$, because it involves approximately a $L = \log_2(m)$ of logical equalities (for example: is $X > Y$?). The following two sections present two new SPITs, the first of which decreases the amount of logical equalities L by taking advantage of a best-fitting criterion, whereas the second is such that no searching steps are required.

Best-Fitting SPIT

Figure 4 plots the **S** vector as a function of its indices for the reference star tracker ($\vartheta_{\text{FOV}} = 30$ deg, $\epsilon = 10$ arcsec, $M_T = 3$). At first glance the **S** vector appears to increase with a linear slope. Therefore, the idea to represent the m points $[i, \mathbf{S}(i)]$ by the best-fitting linear function $\overline{I} = \alpha \overline{S} + \beta$ arises. The least-square fitting is performed by minimizing the following loss function

$$L = \sum_{1}^{m}{}_{i} \{\overline{I}[\mathbf{S}(i)] - i\}^2 = \sum_{1}^{m}{}_{i} [\alpha \mathbf{S}(i) + \beta - i]^2 \tag{6}$$

the solution of which is

$$\begin{Bmatrix} \alpha \\ \beta \end{Bmatrix} = \begin{bmatrix} \Sigma_i [\mathbf{S}(i)]^2 & \Sigma_i \mathbf{S}(i) \\ \Sigma_i \mathbf{S}(i) & m \end{bmatrix}^{-1} \begin{Bmatrix} \Sigma_i i \mathbf{S}(i) \\ m(m+1)/2 \end{Bmatrix} \tag{7}$$

The best-fitting line $\overline{I} = \alpha \overline{S} + \beta$ is a good tool to estimate the indices of the two observed stars spatially separated by the angle ϑ. In fact, $k = \text{int}\{\alpha \cos\vartheta + \beta\}$, where $\text{int}\{z\}$ is the function providing the integer number closest to z, is a good guess of the right element-index of the **S** vector giving the dot product $\cos\vartheta$. This means that the admissible star-pairs (associated with the observed one) is located close to the star-pair $[\mathbf{v}_i \mathbf{v}_j]$ identified by the indices $i = \mathbf{I}(k)$ and $j = \mathbf{J}(k)$, where

$$k = \text{int}\{\alpha \cos\vartheta + \beta\} \tag{8}$$

Let $\mathbf{s}_i$ and $\mathbf{s}_j$ be any two observed stars, separated by the angle ϑ (where $\cos\vartheta = \mathbf{s}_i^{\mathrm{T}} \mathbf{s}_j$), and whose precision is ϵ. The value of $\cos\vartheta$, due to the sensor

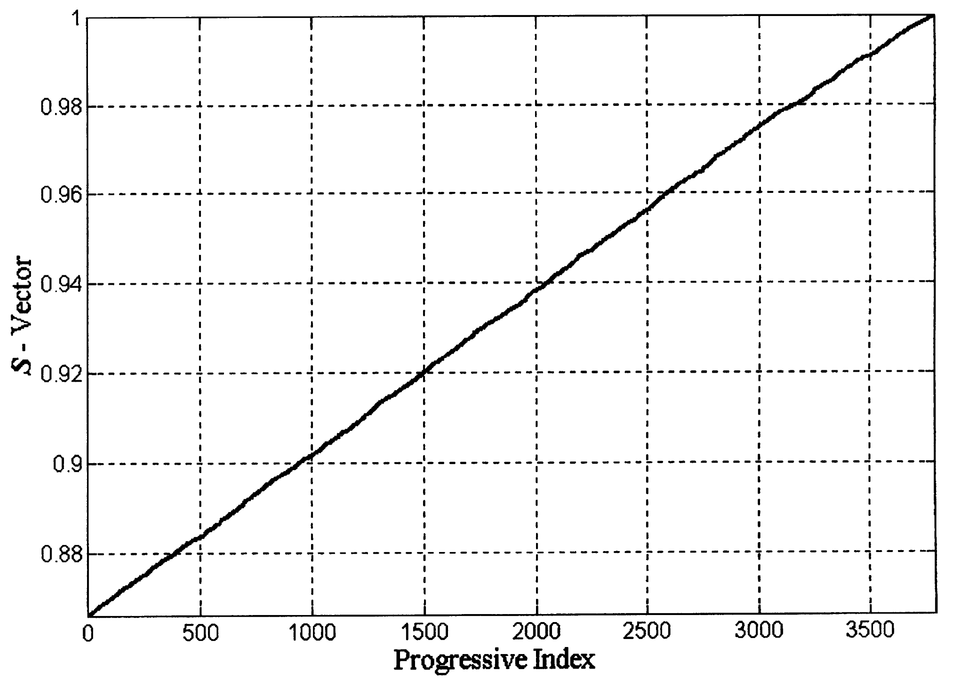

FIG. 4. S-Vector Vs. its Indices.

precision, may vary from the minimum value of $\cos(\vartheta + 2\epsilon)$ to the maximum value of $\cos(\vartheta - 2\epsilon)$. Therefore, the admissible star-pairs, which are located in the **S** vector close to the kth index, are easily found about the kth position provided by equation (8), just by seeking all the star-pairs $\mathbf{v}_{\mathrm{I}(k)}$ and $\mathbf{v}_{\mathrm{J}(k)}$ satisfying the condition

$$\cos(\vartheta + 2\epsilon) \le \mathbf{v}_{\mathrm{I}(k)}^{\mathrm{T}} \mathbf{v}_{\mathrm{J}(k)} \le \cos(\vartheta - 2\epsilon) \tag{9}$$

Obviously, the better the best-fitting function describes the vector **S**, the better k approximates the true index i. Hence, a reduction of $|i - k|$ is achieved by substituting the linear best-fitting function with a more suitable one, which can be, for instance, the polyno-harmonic function

$$\bar{I} = \sum_{0}^{n_p}{}_j\, a_j \bar{S}^j + \sum_{1}^{n_a}{}_j\, A_j \sin[\omega_j \bar{S} + \varphi_j] \tag{10}$$

The $n_p + 1 + 3n_a$ unknown coefficients of this function ($n_p + 1$ coefficients a_j and the $3n_a$ coefficients A_j, ω_j and φ_j) are determined such that $\Sigma_i\{\bar{I}[\mathbf{S}(i)] - i\}^2$ is at a minimum. Particular care must be exercised when performing this best fitting, especially when a high polynomial degree is considered. In order to minimize the numerical errors introduced by the inversion of a poorly conditioned matrix, the CAM best-fitting technique [14] was used. Numerical tests show that the harmonical term in equation (10) does not improve the fitting (FFT analysis does not identify significant frequencies) whereas the polynomial term describes data properly up to the 9th degree when using the CAM best-fitting technique, and lower, using the classic least-square best-fitting technique. Higher degree polynomials imply unacceptable numerical errors. The use of the Legendre orthogonal polynomials was also tested but no practical advantages were found.

Figures 5 and 6 plot the index error $(i - k)$ as a function of the true index i for the two cases of a linear ($n_p = 1$, $n_a = 0$, $(i - k)_{\mathrm{MAX}} = 64$, standard deviation $= 15.87$) and 9th-degree-polynomial ($n_p = 9$, $n_a = 0$, $(i - k)_{\mathrm{MAX}} = 21$, standard deviation $= 6.81$) best-fitting functions. Figures 7 and 8 show the histograms associated with the two considered cases. However, it is to be noted that:

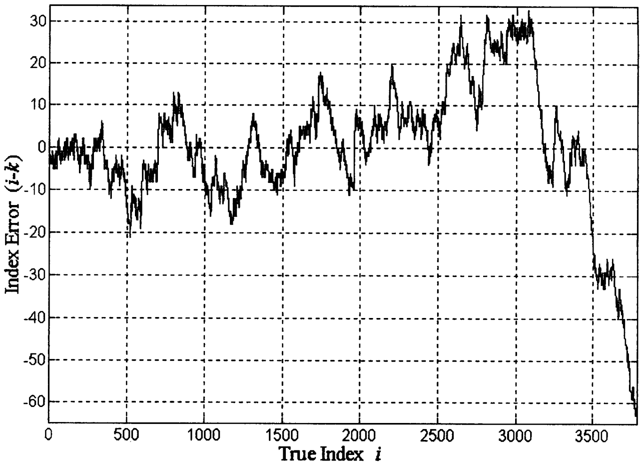

FIG. 5. $(i - k)$ Index Error ($n_p = 1, n_a = 0$).

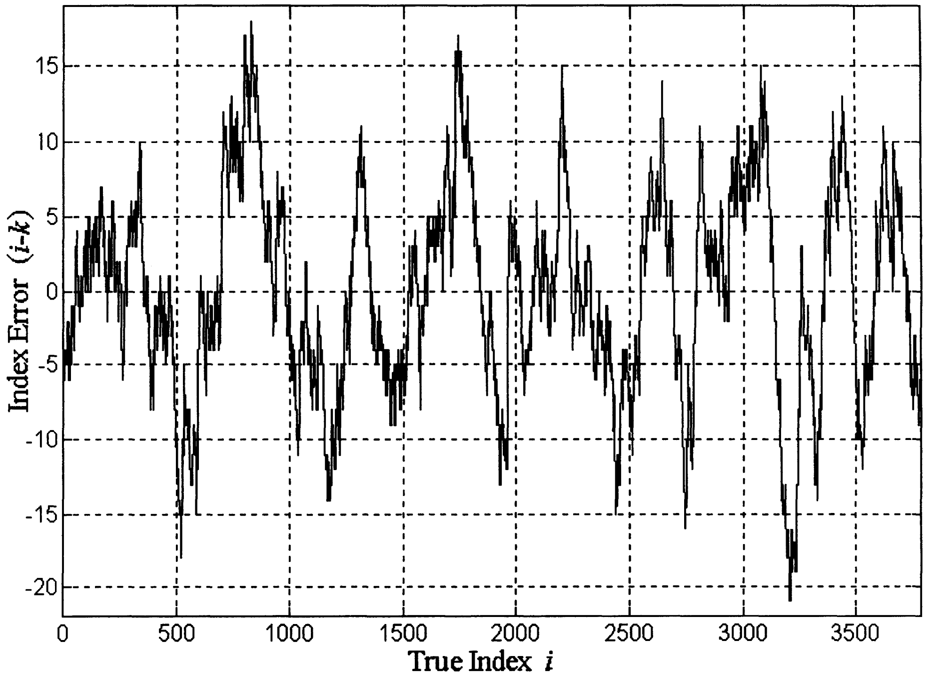

FIG. 6. $(i - k)$ Index Error $(n_p = 9, n_a = 0)$.

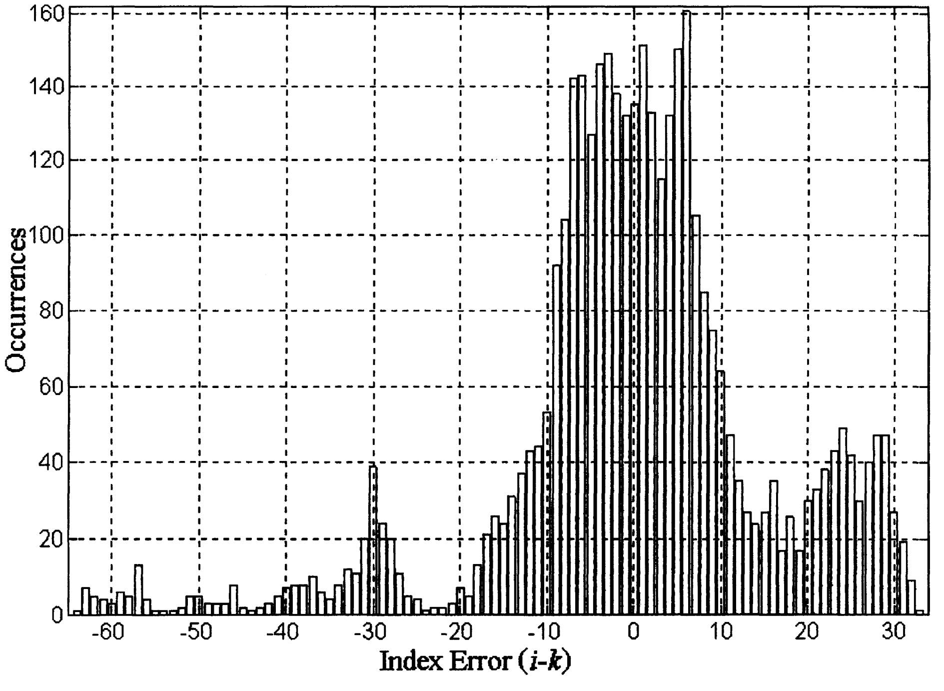

FIG. 7. $(i - k)$ Histogram $(n_p = 1, n_a = 0)$.

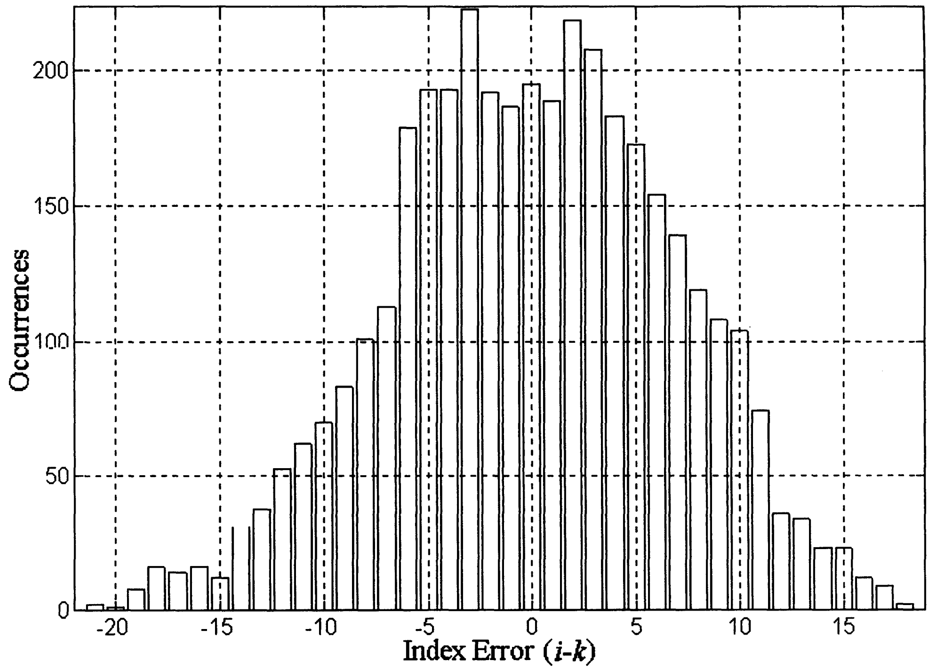

FIG. 8. $(i - k)$ Histogram $(n_p = 9, n_a = 0)$.

1. The $(i - k)$ reduction gain has a meaning only if compared with the range size of the associated admissible star-pair set $[k_{\text{start}}, k_{\text{ends}}]$. The size of $[k_{\text{start}}, k_{\text{end}}]$ mainly depends on the precision ϵ and the set magnitude threshold M_T. In particular, the number of the admissible star-pairs $(k_{\text{end}} - k_{\text{start}} + 1)$ increases with the ϵ increase and for a more sensitive sensor (lower magnitude threshold). For instance, if the average of $(k_{\text{end}} - k_{\text{start}} + 1)$ is 30, the use of a high polynomial degree is unnecessary,
2. The gain of $|i - k|$ obtained by using a higher polynomial degree is also counterbalanced by a higher computational load due to the polynomial computation.

In conclusion, the best polynomial degree choice, associated with the fastest SPIT algorithm, should be identified by numerical tests. The next section describes a better SPIT which does not require a searching phase and hence provides the reason for the Search-Less Algorithm name.

K-Vector SPIT

As already mentioned, a great advantage can be obtained by the following technique, which uses the m-dimensional integer vector $\mathbf{K}$. This vector, which has been devised in order to avoid the searching phase characterizing all the existing SPITs, directly provides the indices of the admissible star-pairs range.

The straight line connecting the two extreme points $[1, \mathbf{S}(1)]$ and $[m, \mathbf{S}(m)]$, has, on average, one element $\mathbf{S}(i)$ for each $D = [\mathbf{S}(m) - \mathbf{S}(1)]/(m - 1)$ step. However, let us consider the slightly steeper line which connects the two points $[1, \mathbf{S}(1) - D/2]$ and $[m, \mathbf{S}(m) + D/2]$. This line, as it will be clear later, assures $\mathbf{K}(1) = 0$ and $\mathbf{K}(m) = m$, and thus simplifies the code by avoiding many index checks. The equation of this line can be written as

$$\cos\vartheta = a_1 k + a_0 \quad \text{where} \quad a_1 = mD/(m-1); a_0 = \mathbf{S}(1) - a_1 - D/2 \tag{11}$$

Starting with $\mathbf{K}(1) = 0$, an integer vector $\mathbf{K}$ is then built as follows

$$\mathbf{K}(k) = l \quad \text{where} \quad \mathbf{S}(l) \le a_1 k + a_0 < \mathbf{S}(l+1),\ k = 2 - m \tag{12}$$

From a practical point of view, the kth element of the $\mathbf{K}$-vector represents the number of elements $\mathbf{S}(i)$ below the value $\cos\vartheta = a_1 k + a_0$. Once this vector has been built, the evaluation of the two indices identifying (in the $\mathbf{S}$ vector) the range of all the possible catalog star-pairs that can match the observed one becomes a straightforward and fast task. The indices associated with these values in the $\mathbf{S}$ vector are simply provided as

$$\begin{aligned} l_{\text{bot}} &= \text{bot}\{[\cos(\vartheta + 2\epsilon) - a_0]/a_1\} \\ l_{\text{top}} &= \text{top}\{[\cos(\vartheta - 2\epsilon) - a_0]/a_1\} \end{aligned} \tag{13}$$

where the function top$\{x\}$ is defined as the larger integer number next to x, and bot$\{x\}$ is the integer number immediately below x. Once l_{bot} and l_{top} are evaluated, it is possible to compute

$$k_{\text{start}} = \mathbf{K}(l_{\text{bot}}) + 1 \qquad \text{and} \qquad k_{\text{end}} = \mathbf{K}(l_{\text{top}}) \tag{14}$$

The knowledge of k_{start} and k_{end} identifies the set of the associated catalog star-pairs (matching with the observed one) as that constituted by all of the $[\mathbf{v}_i \mathbf{v}_j]$ star-pairs, where $i = \mathbf{I}(k)$ and $j = \mathbf{J}(k)$ and

$$k_{\text{start}} \le k \le k_{\text{end}} \tag{15}$$

Equations (13-15) are the equations of the $\mathbf{K}$-vector SPIT. It is to be noted that the $\mathbf{K}$-vector completely substitutes the vector $\mathbf{S}$. This is an additional advantage because $\mathbf{K}$ in an integer vector whereas $\mathbf{S}$ is real. Therefore, a saving on memory storage is obtained when using the $\mathbf{K}$-vector SPIT. For the sake of clarity, Fig. 9 shows the construction of the $\mathbf{K}$-vector for $m = 11$ dot products only. The $\mathbf{K}(i)$ element represents the total number of elements $\mathbf{S}(j)$ satisfying the condition $\mathbf{S}(j) \le a_i i + a_0$, where $j = 1 - m$. It is to be noted that, as shown in this figure, some catalog star-pairs which do not match with the observed star-pair, that is, which do not fall in the range $[\cos(\vartheta + 2\epsilon)\ \cos(\vartheta - 2\epsilon)]$, may occur on the edges of the range $[k_{\text{start}}, k_{\text{end}}]$. For instance, in the example shown in Fig. 9, the elements $\mathbf{S}(5)$ and $\mathbf{S}(6)$ are not admissible, that is, they do not satisfy equation (9). However, the average number of these elements for any star-pair set is only one because, on average, there is only one occurrence of $\mathbf{S}(i)$ for each D step and the elements, whose indices are l_{bot} and l_{top}, have a 50% probability each one of being included within the range $[\cos(\vartheta + 2\epsilon)\ \cos(\vartheta - 2\epsilon)]$.

Both the Best-Fitting and the $\mathbf{K}$-vector SPITs can be implemented in every star identification algorithm. The next section illustrates a new SMIT based on a privileged Reference-Star which identify the observed stars evaluating only $(n-1)$ [instead of all the $n(n-1)/2$] angular-separation combinations.

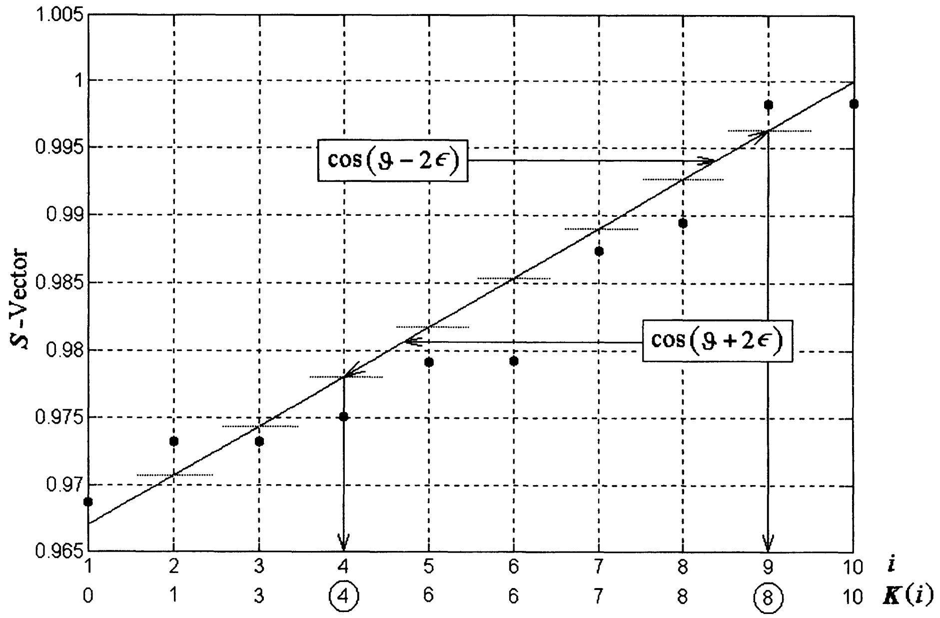

FIG. 9. K-Vector Construction Example.

Reference-Star SMIT

Once the admissible star-pair sets have been calculated, there arises the problem of identifying which stars are the observed ones. This is accomplished here with the proposed Reference-Star method, which: 1) does not use the magnitude information, 2) is able to identify and discard spurious data (spikes, phantom stars), if any, and 3) is fast and reliable.

Let n be the sum of the n_s observed spikes (which could be caused by high-energy cosmic rays, electronic noise, planets, light reflections, et cetera) and the n_t true observed stars $\mathbf{s}_i$, provided by the sensor. Let the rth observed star $\mathbf{s}_r$ be the reference star. Starting from this star, which may be any one (even a spike), the **K**-vector technique is applied to all the star-pairs $[\mathbf{s}_r \mathbf{s}_k]$ (with $k \neq r$). Let $r = 1$ be the starting Reference-Star and let us assume that the **K**-vector SPIT finds, for any observed (r, k) star-pair $(1 \leq k \leq n, k \neq r)$, a number of h admissible star-pairs which can be arranged to form the h-dimensional index vectors $\mathbf{I}_{r,k}$ and $\mathbf{J}_{r,k}$

$$\begin{aligned} \mathbf{I}_{r,k} &\equiv \{i_{r,k,1}\ i_{r,k,2}\ \cdots\ i_{r,k,h-1}\ i_{r,k,h}\}^{\mathrm{T}} \\ \mathbf{J}_{r,k} &\equiv \{j_{r,k,1}\ j_{r,k,2}\ \cdots\ j_{r,k,h-1}\ j_{r,k,h}\}^{\mathrm{T}} \end{aligned} \tag{16}$$

where h is a function of k and r.

Once all of the $(n-1)\,\mathbf{I}_{r,k}$ and $\mathbf{J}_{r,k}$ admissible vectors $(k \neq r)$ have been evaluated, it is possible to arrange them all together to form the overall admissible (for the rth star) index vectors $\mathbf{I}_r$ and $\mathbf{J}_r$

$$\begin{aligned} \mathbf{I}_r &\equiv \{\mathbf{I}_{r,1}^{\mathrm{T}}\ \mathbf{I}_{r,2}^{\mathrm{T}}\ \cdots\ \mathbf{I}_{r,n-1}^{\mathrm{T}}\ \mathbf{I}_{r,n}^{\mathrm{T}}\}^{\mathrm{T}} \\ \mathbf{J}_r &\equiv \{\mathbf{J}_{r,1}^{\mathrm{T}}\ \mathbf{J}_{r,2}^{\mathrm{T}}\ \cdots\ \mathbf{J}_{r,n-1}^{\mathrm{T}}\ \mathbf{J}_{r,n}^{\mathrm{T}}\}^{\mathrm{T}} \end{aligned} \tag{17}$$

The overall $\mathbf{I}_r$ and $\mathbf{J}_r$ admissible vectors contain all of the star-pair indices involving the rth observed star with all the others.

If $\mathbf{s}_r$ is not a spike, then, in the index vector $\mathbf{L}_r = \{\mathbf{I}_r^{\mathrm{T}}, \mathbf{J}_r^{\mathrm{T}}\}^{\mathrm{T}}$, the true index associated with the rth observed star appears $(n_t - 1)$ times, at least. Let us accept a spike presence of up to 25% of n, that is, up to $n_{\mathrm{smax}} \leq \mathrm{bot}\{n/4\}$. This means, for example, that the presence of only one spike is tolerated if the number of observed stars ranges from 4 to 7, two spikes if it ranges from 8 to 11, and so on. Obviously, if $n = 3$, the presence of a spike is unacceptable. Note that a 25% spike presence is vastly superior to the value actually experienced of a few spikes per year.

Based on the following steps/rules, the Reference-Star SMIT is derived:

- If the index vector $\mathbf{L}_{r,k} = \{\mathbf{I}_{r,k}^{\mathrm{T}}, \mathbf{J}_{r,k}^{\mathrm{T}}\}^{\mathrm{T}}$ has no elements, it could mean that: 1) the kth star is a spike, or 2) the rth Reference-Star is a spike, or 3) both stars are spikes. Let n_x be the number of void $\mathbf{L}_{r,k}$ vectors (those having no elements).
- If $n_x > n_{\mathrm{smax}}$, it means that the data are too noisy or the selected Reference-Star is a spike. In this case the index r is increased by $r = r + 1$, and the associated index vectors $\mathbf{I}_r$ and $\mathbf{J}_r$ are evaluated. If $r > n - 3$, the data are declared too noisy and the star identification process is aborted.
- If $n_x \leq n_{\mathrm{smax}}$, then a histogram of the $\mathbf{L}_r$ vector is performed. Let f_1 and f_2, where $f_1 \geq f_2$, be the two highest values of this histogram, which are the occurrences of l_1 and l_2 indices, respectively.
- If $f_1 > 3n/4 - 1$ and $f_2 < 1 + \mathrm{int}\{n/5\}$, then the rth star index is identified as l_1; otherwise, another reference star is chosen and r is increased $(r = r + 1)$.

Once the reference star is identified, it is possible to proceed with the identification of the other observed stars by the analysis of the index vectors $\mathbf{I}_{r,k}$ and $\mathbf{J}_{r,k}$. If l_1 appears once in the $\mathbf{L}_{r,k}$ vector, then the kth star is immediately identified. Let p be the position of l_1 in the vectors $\mathbf{I}_{r,k}$ or $\mathbf{J}_{r,k}$. Hence, the kth star index is identified as l_k, where

$$\begin{aligned} &\text{if} \quad l_1 = \mathbf{I}_{r,k}(p) \quad \text{then} \quad l_k = \mathbf{J}_{r,k}(p) \\ &\text{if} \quad l_1 = \mathbf{J}_{r,k}(p) \quad \text{then} \quad l_k = \mathbf{I}_{r,k}(p) \end{aligned} \tag{18}$$

that is, l_k can be $i_{r,k,p}$ or $j_{r,k,p}$ only, where $1 \leq p \leq h$ and h, which depends on the (r, k) observed stars considered, represents the length of the $\mathbf{I}_{r,k}$ or $\mathbf{J}_{r,k}$ vectors.

Software Structure Description

The proposed overall software structure is shown in the flowchart of Fig. 10. Software programs are currently written in MATLAB [15]. They consist of two main programs: the Ground-run Program (GP) and the On-Board-running Program (OBP).

The GP is carried out only once prior to the spacecraft launch. It is fed by: 1) the star catalog data, 2) the star tracker characteristics (the direction of its boresight axis, the maximum lens aperture ϑ_{FOV} and the instrument precision ϵ), and c) the magnitude threshold M_T. The GP provides the OBP with the $\mathbf{I}, \mathbf{J}$ and $\mathbf{K}$ integer vectors associated with the value of M_T. GP provides the best-fitting

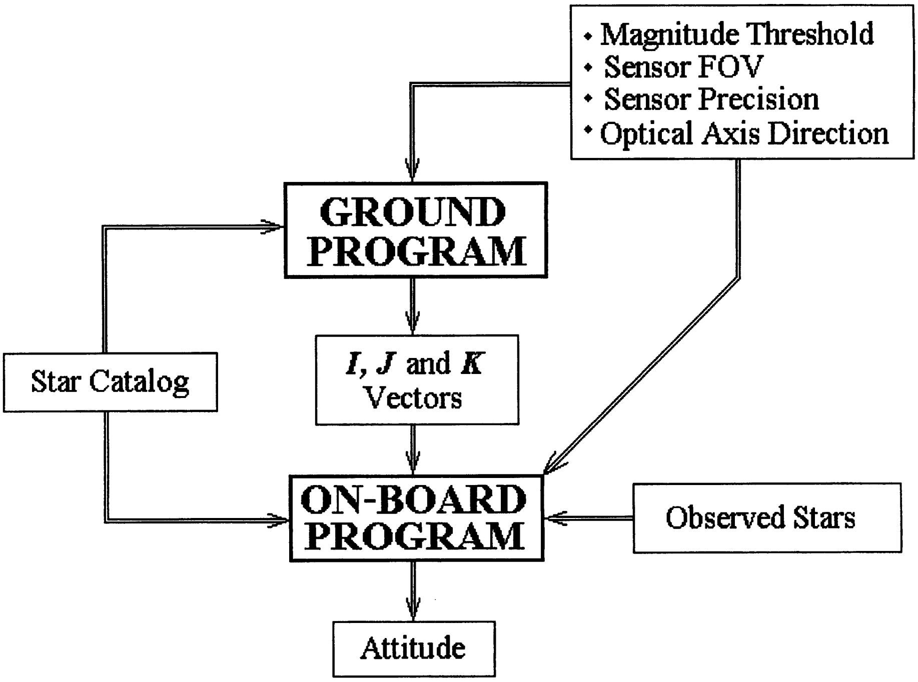

FIG. 10. SLA Software Flow Diagram.

coefficients together with the **I**, **J** and **S** vectors if the **K**-vector is replaced by the Best-Fitting SPIT. The OBP permanently accomplishes the star identification process and the attitude estimation during flight. It is fed by: 1) the star catalog, 2) the directions of the observed stars from the CCD camera and, 3) the **I**, **J** and **K** integer vectors from GP.

In order to attain the maximum computational speed, the **K**-vector SPIT and the *ESOQ* optimal attitude estimation algorithm [1] have been used during numerical tests and are recommended for the following reasons: 1) the **K**-vector SPIT is faster and requires less memory storage than the Best-Fitting SPIT, and 2) the *ESOQ* attitude estimation algorithm is optimal, nonsingular, and the fastest among the existing optimal attitude estimation algorithms.

Numerical Tests

In order to validate the proposed star identification technique, appropriate tests have been carried out. Simulation tests are performed on a 166 MHz Pentium-PC, on a basis of $N = 1000$ random attitudes. The **K**-vector SPIT is chosen. Both noise and an amount of spikes, up to 5% of the observed true stars, have been added to the data. The noise in the star directions is applied by a rigid rotation about a random axis perpendicular to the star direction of a random angle not greater than the sensor precision $\epsilon = 10$ arcsec. Tests are related to a reference star tracker having $\vartheta_{\text{FOV}} = 30$ deg and $M_T = 3$.

Figure 11 represents the speed test results. It shows the cumulative number of floating point operations, obtained by using the MATLAB function "flops" [15], for the complete SLA star identification process as a function of the number n of the observed stars. Figure 12 displays the maximum attitude error δ [16] between

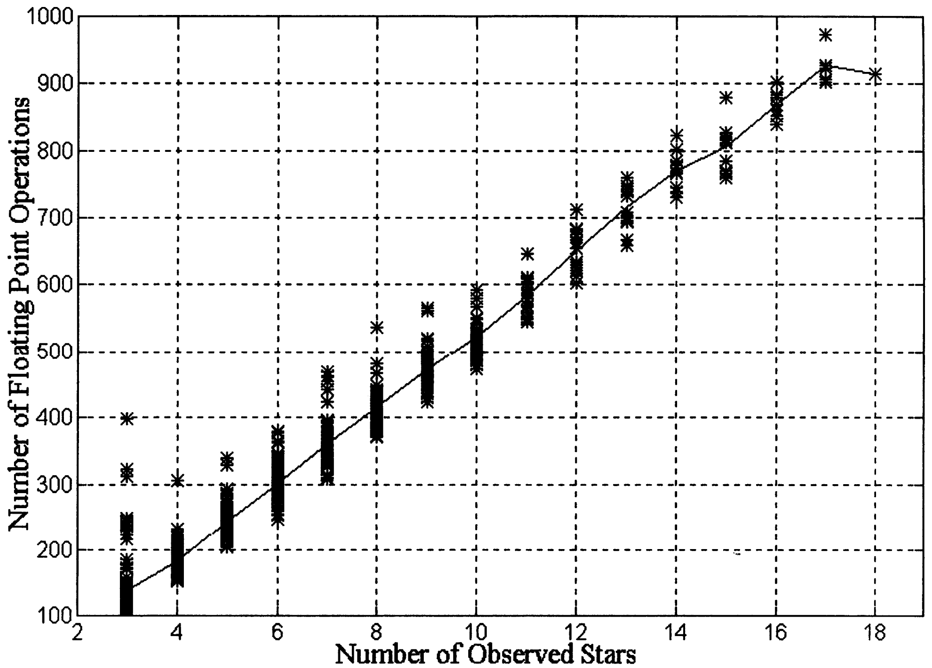

FIG. 11. Speed Test Results.

the true attitude matrix T and estimated attitude matrix A

$$\delta = \cos^{-1}\left(\frac{\text{trace}[TA^{T}] - 1}{2}\right) \tag{19}$$

as a function of n. The average value of the admissible star pair number, that is, the average value of $h = (k_{\text{end}} - k_{\text{start}} + 1)$, is 1.414.

Conclusion

This paper presents the new Search-Less Algorithm (SLA) for star pattern recognition. This algorithm, which does not use the magnitude information, performs the star identification process without a searching phase and without the knowledge of an a priori attitude guess (even though this knowledge would make the algorithm faster). SLA, which is able to identify and to discard spikes (observed true stars, which are missing from the catalog, are treated as spikes) is obtained by combining a Star-Pair Identification Technique (SPIT) and a Star-Matching Identification Technique (SMIT).

Two different SPITs are proposed: the first is based on a best-fitting criterion (which still requires a small searching phase) whereas the second, which allows a search-less direct identification, is based on an appropriate vector of integers, the **K**-vector. The proposed SPITs can be adopted and included in all of the existing star identification processes which use criteria based on either angular separation or triangle patterns. The fast and reliable Reference-Star SMIT, capably of identifying and discarding spikes, is also presented.

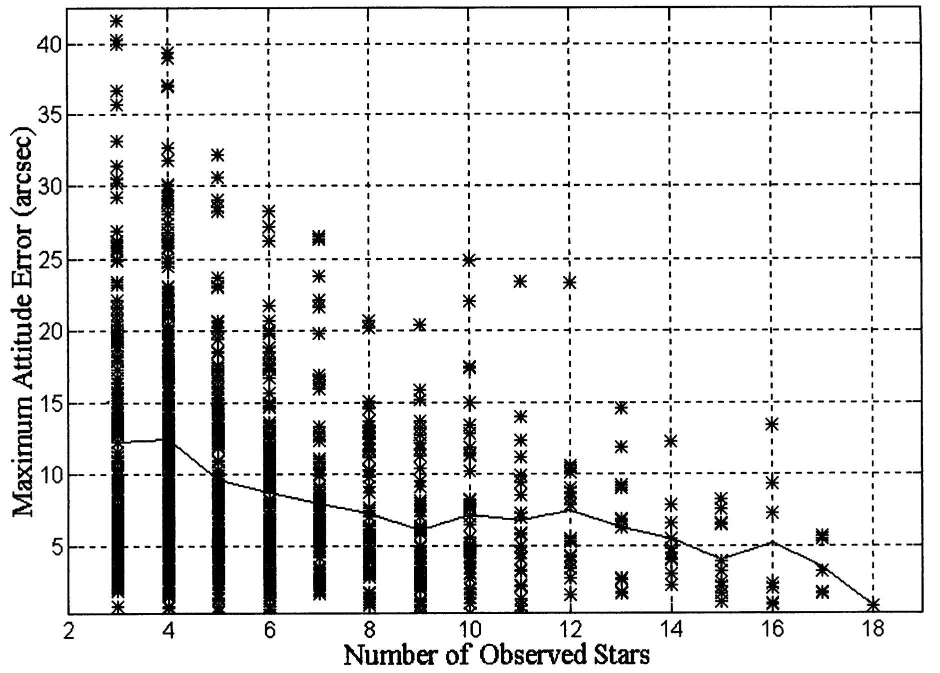

FIG. 12. Accuracy Test Results.

The resulting SLA software, which is constituted by the **K**-vector SPIT and Reference-Star SMIT, was numerically tested and the SLA overall flow diagram was also depicted. Speed tests illustrate, for the reference star sensor, the small number of floating point operations required for the star identification process. Accuracy tests demonstrated, by plotting the maximum attitude error between the true and the estimated attitude (optimally provided by the ESOQ attitude determination algorithm), the SLA reliability as well as the accuracy obtained for the considered reference star sensor.

SLA and ESOQ algorithms constitute a complete attitude determination algorithm which is fast, accurate, and reliable. Therefore, this algorithm can be considered a suitable solution for on-board autonomous attitude determination systems using wide-FOV star sensors.

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